

BobaLink Browser Extension “ Feature Coverage Ledger

Revision: 2 **Last modified:** 2026-06-10T20:45:00Z **Scope:** BobaLink (extension/) “ feature — test-type coverage ledger with a per-feature autonomous-verification class (Â§11.4.25 full-automation-coverage, Â§11.4.52 autonomous-validation). **Authority:** master plan docs/browser_extension/IMPLEMENTATION_PLAN.md (Phase 8, T8.5); live status docs/browser_extension/Status.md (Rev 4).

Â§11.4.6 (no-guessing) / Â§11.4.5 (captured evidence): every “ cell cites a REAL test file backing it. A cell with no real test file is **PENDING** “ never silently marked covered. The e2e row is honestly **OPERATOR_ATTENDED_ONLY** in this headless sandbox: the MV3-load test is real but **SKIPS** (no extension service worker registers “ Â§11.4.3 topology dispatch), so it is not autonomous here.

Ground-truth baseline (this session)

- Full suite: **413 passed / 37 spec files** (`npx vitest run`) under `extension/tests/{unit,perf,stress,chaos,integration,security,e2e}+src/**`.
- E2E (`tests/e2e/extension-loads.spec.ts`) is a real Playwright MV3-load test that **SKIPS-with-reason** in this headless/sandbox environment (no display / unpacked-extension load unsupported); it executes the popup/options assertions only on a headful-capable host. Honest gap per Â§11.4.3, never a faked PASS.
- Challenge: `challenges/extension/detect_and_forward_challenge.sh` drives the real orchestrator + client end-to-end (`detect â†’ POST :7187/api/v1/download`), PASS on captured evidence, mutation-verified (no-op stub â†’ FAIL).
- HelixQA bank: `challenges/helixqa-banks/boba-bobalink.yaml` (symlinked from `submodules/helixqa/banks/boba-bobalink.yaml`) “ `http`: cases require a LIVE merge-service on `:7187`; reports connection failure if down (no fail-open).

Legend

- “ path “ covered by the cited real test file (assertion on a user-observable outcome; fails against a no-op stub).
- **SKIP** “ a real test exists but is operator-gated / topology-gated in this sandbox (e2e MV3 load; live-: 7187 HelixQA).
- **PENDING** “ no real test file yet (tracked in Gaps).
- **N/A** “ test type does not apply to this client-side feature.

Autonomous-verification class (Â§11.4.52)

- **AUTONOMOUS_VERIFIED** “ at least one autonomous path runs headlessly here with captured evidence + PASS/FAIL with no human in the loop.
- **AUTONOMOUS_DESIGNED** “ an autonomous path exists/designed but its primary end-to-end proof is gated on an external dependency not present in this sandbox (live : 7187, headful extension load).
- **OPERATOR_ATTENDED_ONLY** “ the only proof path needs a host with a real display / operator (release blocker until promoted; tracked in Gaps).

- **NOT_APPLICABLE** – feature class does not warrant the path.

Coverage matrix

Feature / flow	unit	integration	security	stress
Magnet detection (BEP-9 parse, scan, highlight)	“ tests/unit/magnet.test.ts, tests/unit/link-scanner.test.ts, tests/unit/text-scanner.test.ts, tests/unit/content.test.ts	“ tests/integration/pipeline.test.ts (STAGE 1)	PENDING	“ tests/parsers.s
.torrent infohash (bencode + SHA-1)	“ tests/unit/torrent-file.test.ts, tests/unit/bencode.test.ts	“ tests/integration/pipeline.test.ts (STAGE 1 .torrent link)	PENDING	“ tests/parsers.s
Cross-scanner dedup (by lowercase infohash)	“ tests/unit/orchestrator.test.ts, tests/unit/scanner-base.test.ts	“ tests/integration/pipeline.test.ts (STAGE 1: magnet— 2 + 1)	PENDING	PENDING
Send + Boba :7187/api/v1/download	“ tests/unit/boba-client.test.ts, tests/unit/api-queue.test.ts	“ tests/integration/pipeline.test.ts (STAGE 2/2b: captured request URL + body)	PENDING	PENDING
Offline FIFO queue (persist, retry, dead-letter, SW-restart)	“ tests/unit/api-queue.test.ts	“ tests/integration/pipeline.test.ts (STAGE 3/3b/3c persist + round-trip + drain)	PENDING	“ tests/queue.str (¥1000 er concurrent)
Popup UI (detected list, Send, Send-All, status)	“ tests/unit/popup.test.ts, tests/all/popup.all.test.ts (9 ARIA)	“	PENDING	N/A
Options UI (7 tabs, save/load config)	“ tests/unit/options.test.ts, tests/all/options.all.test.ts (9 ARIA)	“	“ tests/security/secret-storage.test.ts	N/A
Background routing (SW message router, menus, badge, alarms)	“ tests/unit/background.test.ts, tests/unit/background-token.test.ts	“	PENDING	N/A

Feature / flow	unit	integration	security	stress
Token decrypt-before-send (no embedded key, session passphrase)	“ tests/unit/crypto.test.ts, tests/unit/boba-client-token.test.ts, tests/unit/background-token.test.ts	“ tests/integration/pipeline.test.ts (STAGE 2c forward)	“ tests/security/secret-storage.test.ts, tests/security/no-hardcoded-secret.test.ts	PENDING
Tab-group batch (dedup across group â†’ one POST)	“ tests/unit/tabgroups.test.ts, tests/unit/background.test.ts (MENU_SEND_GROUP)	”	“ tests/security/manifest-least-privilege.test.ts (tabGroups minimal, no tabs)	PENDING
il8n (locale-key completeness)	“ tests/unit/locale.test.ts	”	N/A	N/A
Manifest least-privilege / CSP (cross-cutting)	”	”	“ tests/security/manifest-least-privilege.test.ts, tests/security/csp.test.ts	N/A

Gaps (highest-priority missing coverage ” tracked follow-ups)

- 1. Live : 7187 integration / end-to-end Challenge (Phase 4).** The send-flow integration substitutes the network boundary (fetchImpl stub) and the HelixQA bank (challenges/helixqa-banks/boba-bobalink.yaml) SKIPS when the merge-service is down. PENDING: a require_backend(7187) integration that actually POSTs to a running backend and asserts the torrent appears in qBittorrent (detect â†’ send â†’ server-side add), plus BE-1 (CORS) / BE-2 (. torrent multipart upload) backend work. Promotes "Send â†’ :7187" from AUTONOMOUS_DESIGNED â†’ AUTONOMOUS_VERIFIED.
- 2. E2E real-extension load (Phase 8).** tests/e2e/extension-loads.spec.ts is real but OPERATOR_ATTENDED_ONLY in this sandbox (no MV3 unpacked-extension load; no display). PENDING: run on a headful-capable host so the popup/options/SW assertions actually execute, and add a real multi-tab tab-group e2e (Phase 5) and an offline-queue-durability e2e (SW termination).
- 3. . torrent infohash end-to-end Challenge.** The detectâ†’ forward Challenge covers the magnet path; the . torrent-file forward path has no challenge- level captured-evidence run yet. PENDING.
- 4. Additional locales + deeper-WCAG + theme evidence (Phase 6).** locale.test.ts guards only the en catalog; the plan targets 8 locales. Structural ARIA ally IS now covered (tests/ally/popup.ally.test.ts + options.ally.test.ts ” 18 tests: roles / accessible-names / tablistâ†’i,Žtabpanel / live-regions, mutation-proven). PENDING: 7 more locale catalogs + a locale-switch render test, the deeper WCAG 2.1 AA

checks a static HTML parse cannot cover (contrast ratios, focus order, keyboard traversal), and dark/light theme-switch evidence.

5. **Security depth (Phase 7).** Present: least-privilege manifest, CSP, no-hardcoded-secret, secret-storage. PENDING: Â§11.4.10.A pre-store credential-leak audit + pre-commit grep gate, a cred-leak-grep Challenge, and a fuller pen-test suite (magnet/dn XSS sanitization, sender.id/sender.url validation, log-redaction proof).
6. **Stress/chaos breadth beyond parsers + queue.** Â§11.4.85 stress+chaos exist for parsers and the offline queue. PENDING: stress/chaos for the scanner orchestrator (10k-link page, frame-budget), the BobaClient rate-limiter under API flood, and tab-group batching under many tabs.
7. **Per-store packaging + manual gate (Phase 9).** No extension/ci-ext.sh manual gate, per-store zips, bundle-size assertion, or Â§11.4.65 doc siblings yet. PENDING (NO CI/CD â€” manual only).